ED Management of Mild Diabetic Acidosis with Subcutaneous Insulin

Purpose: To improve the ED flow and management of patients with known diabetes who do not require insulin drips

Patient presentation: Known diabetes treated with insulin via injections coming in with hyperglycemia

1. Check fingerstick glucose and urine ketones, lytes, VBG
2. Administer 10 mL/kg NS bolus
3. If initial pH ≤ 7.25 and/or bicarb ≤ 15 and not tolerating POs/lethargic → proceed with DKA protocol
4. If initial VBG pH ≥7.26, proceed with initial ED management below.
5. Repeat blood glucose after NS bolus.
6. If patient not in DKA, repeat 10 ml/kg bolus
7. Verify home insulin regimen with patient
8. Administer short acting insulin (Humalog) for hyperglycemic correction
    a. If home correction factor is known:
        i. Negative urine ketones: Administer Humalog per home correction
        ii. Small urine ketones: Administer 1.1x home correction
        iii. Moderate urine ketones: Administer 1.25x home correction
        iv. Large urine ketones: Administer 1.5x home correction
    b. If home correction factor unknown:
        i. Negative urine ketones
            1. BG 251-400 mg/dL → Humalog 0.05 U/kg
            2. BG > 400 mg/dL → Humalog 0.1 unit/kg
        ii. Small urine ketones
            1. BG 180-250 mg/dL → Humalog 0.05 unit/kg
            2. BG 251-400 mg/dL → Humalog 0.05 unit/kg
            3. BG > 400 mg/dL → Humalog 0.1 unit/kg
        iii. Moderate urine ketones
            1. BG 180-250 mg/dL → Humalog 0.05 unit/kg
            2. BG 251-400 mg/dL → Humalog 0.1 unit/kg
            3. BG > 400 mg/dL → Humalog 0.1 unit/kg → repeat dose after 2 hours if ketones do not decrease
            4. Large urine ketones
        iv. BG 180-250 mg/dL → Humalog 0.1 unit/kg
            1. BG >250 mg/dL → Humalog 0.1 unit/kg → repeat dose after 2 hours if ketones do not decrease
9. If patient BG decreases, able to tolerate oral hydration with no emesis, and family able to continue close monitoring, ok to discharge to home. Instruct patient to check blood glucose and administer insulin for correction every 3 hours at home until BG normalizes. Encourage oral hydration.
10. Notify on-call pediatric endocrinologist

Case Scenario #2: Patient with diabetes treated with insulin via pump therapy

Follow above steps and administer correction via manual injections with syringes

If patient has + ketones, please have patient change pump site